

MR AHMED ABD EL-MOTAAL**Lesson One: Proportion**

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SECTION 1: CONCEPT & PRACTICE

Student Name: _____ **Class / Group:** _____ **Date:** ____ / ____ / 2026 **Evaluation Score:** [____ / 10]

🎓 Pre-Study: Check Your Readiness

Review essential prerequisite concepts: simplifying fractions, ratio equivalence, and rates before studying proportion.

🕒 Prerequisite Review**1. Simplifying Fractions to Lowest Terms**

Divide the numerator and denominator by their greatest common divisor (GCD):

- $\frac{12}{18} = \frac{12 \div 6}{18 \div 6} = \frac{2}{3}$
- Ratios with identical irreducible fractions represent equivalent quantities.

**2. Understanding Ratios & Rates**

A ratio compares quantities of the same unit, while a rate compares different units:

- **Ratio:** 4 pens to 6 pens $\rightarrow 4 : 6 = 2 : 3$
- **Rate:** 150 kilometers in 2 hours $\rightarrow 75 \text{ km/h}$

**3. Cross-Multiplication Principle**

If two fractions are equal $\frac{a}{b} = \frac{c}{d}$, their cross products are equal:

$$a \times d = b \times c$$

If $\frac{x}{8} = \frac{3}{4}$, then $4x = 24 \implies x = 6$.

☰ Diagnostic Check: Test Your Readiness

QUESTION 1 OF 3

1. Which ratio is equivalent to 3 : 5?

A 12 : 15

B 6 : 8

C 9 : 15

D 5 : 3

QUESTION 2 OF 3

2. If $\frac{x}{6} = \frac{10}{15}$, what is the value of x ?

A 3

B 5

C 4

D 6

QUESTION 3 OF 3

3. If a vehicle travels 180 km in 3 hours, what is its uniform speed rate?

A 50 km/h

B 55 km/h

C 45 km/h

D 60 km/h

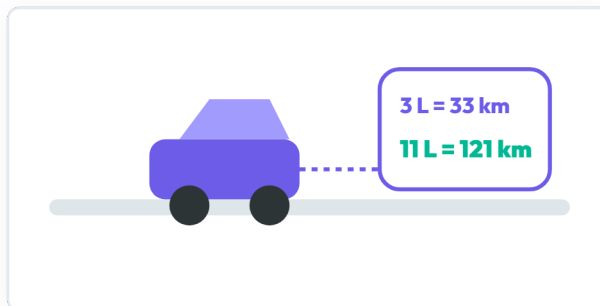


Real-World Connections (Mathematics in Everyday Life)

Precision Plots & Modeling

🚗 Automotive & Travel • Car Petrol & Highway Mileage**🌐 STEM Real-World Application**

In automotive engineering and journey planning, fuel consumption is directly proportional to distance traveled at uniform cruise speeds.

**💡 Mathematical Connection & STEM Analysis:****Mathematical Proportional Modeling:**

Setting up the rate proportion $\frac{x}{121} = \frac{3}{33}$ yields $33x = 363 \implies x = 11$ liters. Modern vehicle dashboard telemetry continuously evaluates this direct proportion to calculate real-time driving range.

🍰 Culinary Chemistry • Smart Bakery Recipe Scaling**🌐 STEM Real-World Application**

Culinary chemistry and commercial bakery operations rely on exact proportional scaling to ensure recipe consistency from small batches to banquet orders.

**💡 Mathematical Connection & STEM Analysis:****Proportional Scaling Formula:**

With a baseline ratio of $\frac{5 \text{ cups}}{3 \text{ cakes}}$, producing 15 cakes scales the ingredients by a factor of 5: $S = 15 \times \frac{5}{3} = 25$ cups of sugar.

🌙 Space Physics & Gravitation • Earth vs. Moon Gravity Ratio

🌐 STEM Real-World Application

Astrophysics and planetary exploration compare gravitational acceleration across celestial bodies through linear proportional relationships.



💡 Mathematical Connection & STEM Analysis:

Gravitational Proportionality:

The lunar surface gravity is directly proportional to Earth gravity by the constant ratio $k = \frac{15}{90} = \frac{1}{6}$. A probe weighing 60 N on Earth weighs exactly $60 \times \frac{1}{6} = 10$ Newtons on the Moon.

📌 Essential Curriculum Standards

Definitions & Standard Forms

1

Extremes & Means Structure

In $a : b = c : d$, a and d are the Extremes; b and c are the Means.

2

Cross Multiplication Property

Product of Extremes = Product of Means: $a \times d = b \times c$. If $ad = bc$, then a, b, c, d are proportional.

3

Simplest Form Equivalence

Two ratios form a proportion if both reduce to the same irreducible fraction (e.g. $\frac{8}{10} = \frac{4}{5} = \frac{12}{15}$).

4

Zero Denominator Rule

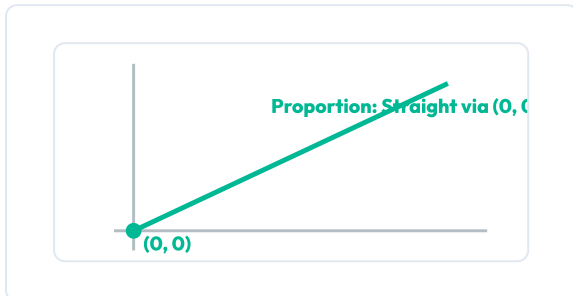
In any proportion $\frac{a}{b} = \frac{c}{d}$, denominators must be non-zero ($b \neq 0, d \neq 0$).



Special Cases, Graphical Models & Geometric Properties

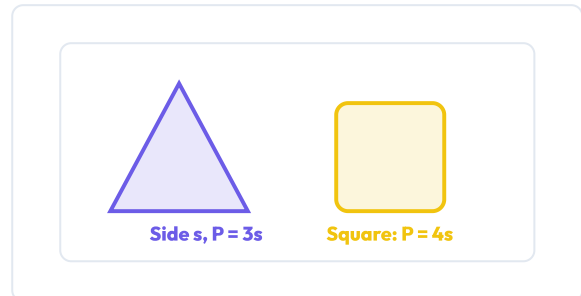
Geometric Analysis

1. Graphical Representation & Origin Test



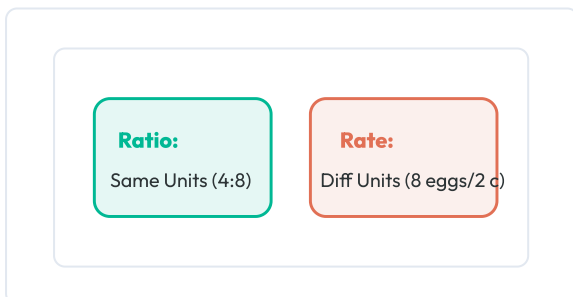
- Points form $(0,0) \Rightarrow$ **Proportional** ($y = kx$).
a straight line passing through
- Points do NOT pass $(0,0)$ or are curved $\Rightarrow$ **Not Proportional**.

2. Geometric Perimeter Proportions



- **Equilateral Triangles:** $\frac{\text{Perimeter}}{\text{Side}} = \frac{3s}{s} = 3$ (Constant $\Rightarrow$ Proportion!).
ratio
- **Squares:** $\frac{\text{Perimeter}}{\text{Side}} = \frac{4s}{s} = 4$ (Constant $\Rightarrow$ Proportion!).
ratio

3. Rate Equivalence vs Ratio



- **Ratio:** Comparison of two quantities of the same kind and same units.
- **Rate:** Comparison of two quantities of different kinds/units.



Idea 1: Testing Proportions via Cross Products & Simplification

Golden Rule: Cross multiply ($a \times d$) and ($b \times c$). If the two products are identical, the pair represents a proportion. If they differ by even 1, it is NOT a proportion!

Solved Example 1.1 • Non-Proportion Case

iPad Apple Pencil Workspace

Determine whether the pair of ratios $\frac{3}{8}$ and $\frac{6}{10}$ represents a proportion.

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Solved Example 1.2 • Verified Proportion Case

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Determine whether the pair of ratios $\frac{8}{10}$ and $\frac{12}{15}$ represents a proportion.

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 Try It Yourself 1 (Your Turn)

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Determine which of the following pairs of ratios represents a proportion:

1) $\frac{5}{7}$ and $\frac{10}{14}$ 2) $\frac{4}{12}$ and $\frac{3}{9}$

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Concept Flashcard & Strategy Guide

Idea 2: Solving for Unknowns in Proportions (Linear & Binomial Terms)



Strategy Guide: When an unknown is inside a binomial like $(x - 3)$ or has a coefficient like $2x$, isolate the group first using cross-multiplication: $x - 3 = \frac{b \times c}{d}$, then solve for x by adding 3!

Solved Example 2.1 • Direct Linear Variable

iPad Apple Pencil Workspace

Solve the following proportions for x :

1) $\frac{4}{12} = \frac{20}{x}$ 2) $\frac{4}{7} = \frac{x}{35}$

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Solved Example 2.2 • Binomial & Coefficient Variables

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Solve the following proportions for x :

1) $\frac{3}{x-3} = \frac{12}{8}$ 2) $\frac{40}{2x} = \frac{5}{7}$

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 Try It Yourself 2 (Your Turn)

iPad Apple Pencil Workspace

Find the value of x in each of the following:

1) $\frac{x}{32} = \frac{9}{36}$ 2) $\frac{8}{x-1} = \frac{16}{10}$

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Concept Flashcard & Strategy Guide

Idea 3: Graphical Coordinate Representations & Proportional Tables



Origin Line Rule: A table of values represents a proportion if and only if $\frac{y}{x} = k$ (a constant unit rate) for all pairs, meaning their plotted points lie on a straight line that passes directly through $(0, 0)$.

 Solved Example 3.1 • Distance vs Time Table

iPad Apple Pencil Workspace

A car covers distances according to the table: (1 hr, 70 km), (2 hr, 140 km), (3 hr, 210 km), (4 hr, 280 km). Determine whether the relationship represents a proportion.

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 Solved Example 3.2 • Non-Proportional Table

iPad Apple Pencil Workspace

Another vehicle logs: (1 hr, 80 km), (2 hr, 100 km), (3 hr, 190 km), (4 hr, 260 km). Determine if it represents a proportion.

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 Try It Yourself 3 (Your Turn)

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Adam types pages on a computer: Time (1 hr $\rightarrow$ 3 p), (2 hr $\rightarrow$ 6 p), (3 hr $\rightarrow$ 9 p), (4 hr $\rightarrow$ 21 p). Determine whether the number of pages is proportional to time.

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**Self-Assessment Reflection**

How confident do you feel with Lesson One: Proportion? Click your current level:


Need Practice


Getting It


Good Understanding


Mastered Confidently

Mr Ahmed Abd El-Motaal

Math teacher and content creator

Empowering middle and high school students with engaging, interactive, visual mathematics. Dedicated to

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